

Mid-term 1: 3D Printing in a Chocolate Factory

(Original problems: do not repost anywhere without permission)

(Recall the honor code & trust your own instincts & knowledge)

Marked out of 70; contribution to final grade varies.

After four years of chemical engineering education, you find yourself being interviewed by a major chocolate producer for a dream job with a dream salary at a dream location. You have no idea what to expect in your interview, and you go in, thinking, I know what I know, and I will do as well as I can.

A series of questions follow. The hiring engineer has to take 100 interviews in a week, and would like to get quick, crisp answers.

- (a) What are the definitions and S. I. units for $\underline{e}$ (combined energy flux), h (heat transfer coefficient), work flux, and Nu (Nusselt number)? (Correct formula will earn you complete credit for the definition). (16)

$$\underline{e} = \left(\frac{1}{2} \rho v^2 + \rho \hat{H} \right) v + [\underline{\gamma} - v] + q \quad \left[\text{W/m}^2 \cdot \text{s} \right]$$

$$h = \frac{q}{\Delta T}$$

$$h = \left[\frac{\text{J}}{\text{m}^2 \cdot \text{K}} \right]$$

$$\text{work flux} = \frac{\text{work}}{\text{area} \cdot \text{time}}$$

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$$\left[\frac{\text{J}}{\text{m}^2 \cdot \text{s}} \right]$$

$$Nu = \frac{2D}{\nu}$$

(dimensionless)

Name

- 6) neglect viscous dissipation
7) ^{heat} conduction $\ll$ ^{heat} convection
8)

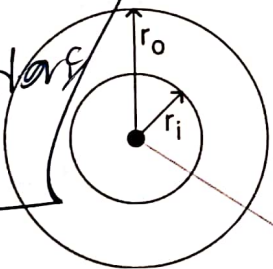
(b) We wish to make hollow chocolate tubes by extrusion. Assume the viscosity, thermal conductivity and heat transfer coefficient are independent of temperature, and presume chocolate melt is a Newtonian fluid.

Assume chocolate is pumped through an annulus region with a constant flow rate with mean velocity u , wherein the outside side of the tube at r_o is exposed to a constant heat flux of q_o at steady state while the stagnant air within the tube of inner radius r_i stays at room temperature T_o . Provide the equations (in dimensional form with applicable boundary conditions or BCs) that need to be solved for determining the temperature distribution in the region between r_i and r_o in steady state.

(HINT: Refer to Appendix B; write down only the relevant terms from the equations of change, and state all your assumptions, postulates and justifications. It is expected that you will identify which terms can be neglected and why, and show all the steps needed for arriving at the final set of equations and BCs. You will be solving for the temperature distribution in the take home version of the exam, here we need only the equations and

BCs). (20)

Please see App B. for solutions



- 1) constant are all physical properties
- 2) $v_z = v_z(r)$ and $v_r, v_\theta = 0$
- 3) steady state
- 4) Newtonian fluid
- 5) $P = P(z)$ and $T(z, r) = T$

CONTINUITY: App B §B.4 cylindrical coordinates
 $0=0$

MOTION: App B. Eq. B.6-6 only since v_θ and $v_r = 0$

$$\frac{\partial P}{\partial z} = \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) \right] + \rho g_z$$

ENERGY: Eq. B.9-2' in App B.

$$\rho \hat{C}_p \left(v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \right]$$

B.C #1's

$$3. C \# 1 \quad z = 0$$

$$P = P_0$$

$$3. C \# 2 \quad z = L$$

$$P = P_L$$

$$3. C \# 3 \quad r = r_i$$

$$T = T_0$$

$$3. C \# 4 \quad r = R$$

$$\varphi_0 = \varphi_r$$

(c) Let us consider a one-dimensional free convection, heat transfer problem in direction x . Let the velocity be v , temperature be T , and let's assume all properties of the fluid are known, and A

and N are constant. For $T = A + 4t + 6\sin x^2 + Nxt$. Compute $\frac{DT}{Dt}$. (10)

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z}$$

— only in x -direction

$$\frac{\partial T}{\partial t} = 4 + Nx$$

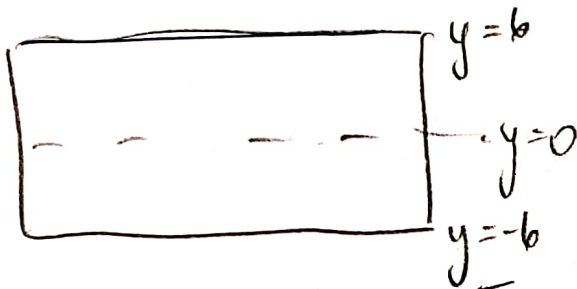
$$\frac{\partial T}{\partial x} = 2 \times 6 \cdot \cos x^2 + Nt$$

$$\frac{DT}{Dt} = (4 + Nx) + v_x (Nt + 6 \cdot 2x \cos x^2)$$

10

- (d) Imagine that a slab of chocolate occupies the space between $y = -b$ and $y = +b$ and is initially at a temperature T_0 . At time $t = 0$, the surfaces at $y = \pm b$ are suddenly raised to T_1 and maintained there. Find $T(y, t)$. State your assumptions clearly. For this problem providing the non-dimensionalized equations and ICs/BCs are worth 20 points, so do not spend all your time on deriving the complete solution (24)

(Please note: 10 points for Equations & Assumptions/Postulates + 10 Initial Conditions and/or Boundary Conditions + 4 Solution).



$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial y^2}$$

Energy balance on a slab solid and Fourier's law provides

Assumptions:

- $T(y, t)$
- no velocity dependence
- all physical properties are constant
- slab is a solid

B.C.#1 $t=0, T=T_0$

B.C.#2 $y = \pm b, T=T_1$

$$\Theta = \frac{T - T_0}{T_1 - T_0} \quad \eta = \frac{y}{b} \quad \tau = \frac{\alpha t}{b^2}$$

$$\frac{\partial \Theta}{\partial \tau} = \frac{\partial^2 \Theta}{\partial \eta^2}$$

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B.C.#3 $\tau=0, \Theta=1$

B.C.#4 $\eta = \pm 1, \Theta=0$

$$\Theta(\eta, \tau) = f(\eta) \cdot g(\tau) = -C^2$$

separation of variables

$$\frac{1}{f} \frac{\partial^2 f}{\partial \eta^2} = \frac{1}{g} \frac{\partial^2 g}{\partial \tau^2} = -C^2$$

↓ App.C. ↓ App.C.

sol. $\Rightarrow f = C_1 \cos C \eta + C_2 \sin C \eta$

sol.

$$A \exp(c^2 \tau)$$

is symmetry

$$\rightarrow \theta(\eta, \tau) = \theta(-\eta, \tau)$$

$$\Rightarrow C_2 = 0$$

since

$$\sin \eta \neq \sin(-\eta)$$

$$3.C \neq 1 \Rightarrow C_1 \cos \eta = 0$$

but

$$C_1 \neq 0$$

$$\Rightarrow \left[C = -\left(n + \frac{1}{2}\right) \pi \eta \right]$$

Then,

$$\theta = A \exp(c^2 \tau) [\cos c \eta]$$

← plug into

$$\theta = A \exp\left(-\left(n + \frac{1}{2}\right)^2 \pi^2 \tau\right) \left[\cos\left(-\left(n + \frac{1}{2}\right) \pi \eta\right) \right]$$

$$\theta = A_n D_n \exp\left(-\left(n + \frac{1}{2}\right)^2 \pi^2 \tau\right) \left[\cos\left(n + \frac{1}{2}\right) \pi \eta \right]$$

$$A_n D_n + A_{-(n+1)} + B_{-(n+1)}$$

$$D_n =$$

$$\text{TRICK } \int_{-1}^1 \cos\left(m + \frac{1}{2}\right) \pi \eta \Rightarrow \int_{-1}^1 \cos\left(m + \frac{1}{2}\right) \pi \eta \cdot \cos\left(n + \frac{1}{2}\right) \pi \eta d\eta$$

$$\left(\begin{array}{l} \text{for all } m \neq n \\ \text{is zero} \\ \text{for all } m = n \end{array} \right) D_m = \frac{2(-1)^m}{\left(m + \frac{1}{2}\right) \pi}$$

←

$$\theta = \frac{T - T_0}{T_1 - T_0} = 2 \sum_{n=0}^{\infty} \frac{2(-1)^n}{\left(m + \frac{1}{2}\right) \pi} \exp\left(-\left(n + \frac{1}{2}\right)^2 \pi^2 \frac{\alpha t}{l^2}\right) \left(\cos\left(n + \frac{1}{2}\right) \frac{\pi y}{l} \right)$$